Solving multi-step inequalities

1

a $x \geq 14$	**b** $x \geq 0$	**c** $x < 5$	**d** $x > -8$
e $x > 2$	**f** $x < -13$	**g** $x \leq -3$	**h** $x \leq 0$
i $x \leq -3$	**j** $x > 12$	**k** $x \leq 16$	**l** $x \geq -24$
m $x \geq 4$	**n** $x < 19$	**o** $x < -21$	**p** $x < -14$
q $x < -2$	**r** $x \geq -1$	**s** $x \leq 7$	**t** $x > 4$
u $x \leq 45$	**v** $x > 40$	**w** $x \leq 3$	**x** $x > 27$

2

a $x > 2$ **b** $x \geq -12$ **c** $x > 8$ **d** $x > \frac{150}{89}$

e $x > \frac{5}{4}$

3

a $x > 0$

b

i No **ii** Yes **iii** Yes **iv** Yes

4

a $x \leq 6$

b

i Yes **ii** Yes **iii** Yes **iv** No

5

a $x > \frac{7}{5}$

b

i No **ii** No **iii** Yes **iv** Yes

6

a

i $x < 4$

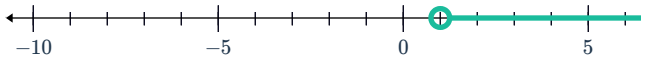
ii



b

i $x > 1$

ii



c

i $x < 0$

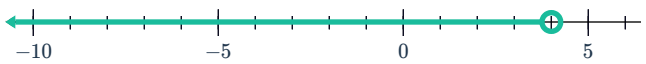
ii



d

i $x < 4$

ii



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e

i $x \geq -1$

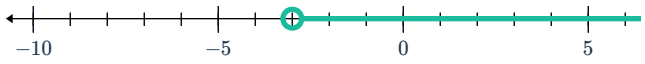
ii



f

i $x > -3$

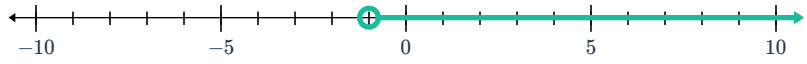
ii



g

i $x > -1$

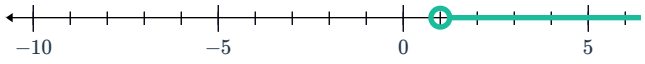
ii



h

i $x > 1$

ii



i

i $x \geq 8$

ii



j

i $x \leq 6$

ii



k

i $x \geq -6$

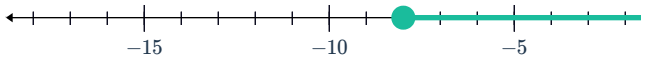
ii



l

i $x \geq -8$

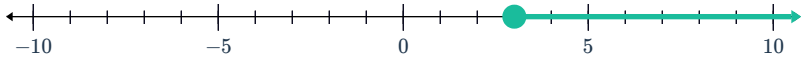
ii



m

i $x \geq 3$

ii



7 Step 1 is incorrect. At this step she incorrectly combined -3 and -2.

8 $x \geq 3$

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9

a $p \leq 5$

b $p = 5$

10 $p \leq 16.75$

11 $A \geq 65.40$

12 $x \geq 66$

13 $W \leq 8$

14

a $x \leq 1630.10$

b $\$1630.10$

15

a $n > \frac{40}{3}$

b 14 rockmelons

16

a $h \geq 61.4$

b 62

17

a $B \leq 3.4$

b 3

18 Lachlan, James

19

a $x + 650 \geq 950$

b $\$960$

c Yes

20

a $10\,060$ lb

b No

Additional questions

21 $3 \leq x \leq 7$ includes both end points, whereas $0 \leq x < 4$ only includes the left hand end point. $0 \leq x < 4$ includes the number 0 and every number up to 4 but not including 4. $3 \leq x \leq 7$ includes the number 3 and every number up to 7 including 7

22

a x can be equal to 29 or any number greater than 29.

b x can be equal to any number less than -29, not including 29.

23

a $f > 2m$

b

i No

ii Yes

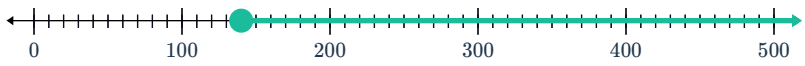
iii No

iv No

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24 **a** $x + 410 \geq 550$

b



25 **a** $w \geq 8.95$

b

i No, washing exactly 9 windows would generate $\$18$, which with the $\$22.10$ he already has is enough money to afford the equipment.

ii No, washing 8 windows would only generate $\$16$ which would give him a total of $\$38.10$, which is not enough money to afford the equipment.

iii No, he gets paid for washing the whole window, not parts.

iv Yes, if Ryan washed 9 windows he would earn enough to buy the equipment. If he washed more than 9 windows, he would have more than enough money.

26 The minimum should be at -9 instead of -8 and the dot at that point should be filled as -9 is included in the solution.

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